QUESTION 1

What is an embedded system?

Answer is Wrong **Score: 0 / 1**

Options

A. A general-purpose computer for running multiple software applications

B. A web-based application for remote access

Expected

C. A computer system designed to perform a dedicated function within a larger system

Wrong

D. A type of gaming software for embedded controllers

FEEDBACK

Solution : An embedded system is a computer system that does a particular task inside a machine or larger electrical system. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=96

QUESTION 2

What is meant by real-time operation in embedded systems?

Not Attempted **Score: 0 / 1**

Options

A. The ability to install and run any software application

B. The ability to play games and stream videos

Expected

C. The ability to perform a specific task repeatedly and respond to inputs within a strict time limit

D. The ability to support multiple users at once

FEEDBACK

Solution : An embedded system performs one specific task repeatedly, while a general-purpose computer can run many different programs.
Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=289

QUESTION 3

Which one of these is essential in building an embedded system?

Not Attempted **Score: 0 / 1**

Options

A. Database server

B. Cloud storage

Expected

C. Microcontroller

D. Laser printer

FEEDBACK

Solution : A microcontroller is the central unit that controls sensors and actuators in embedded systems. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=393

QUESTION 4

What are common constraints in embedded systems?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Limited memory

B. Small physical space

Expected

C. Unlimited power supply

Expected

D. Limited interaction options

FEEDBACK

Solution : An embedded system performs one specific task repeatedly, while a general-purpose computer can run many different programs.

Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=1919

QUESTION 5

Which factors must be considered in embedded system design?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Microcontroller selection

Expected

B. Actuator type

C. Operating system theme

Expected

D. Sensor choice

FEEDBACK

Solution : System-level design includes choosing the right microcontroller, sensors, actuators, and more. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=501

QUESTION 6

Which statements are true about firmware in embedded systems?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Firmware is software tied closely to hardware

B. Firmware is easy to update anytime

C. Firmware updates change fixed device behavior

Expected

D. Firmware is downloaded from Play Store

FEEDBACK

Solution : An embedded system performs one specific task repeatedly, while a general-purpose computer can run many different programs.

Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=1508

QUESTION 7

What communication advances support embedded systems today?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Low-power wireless technologies

B. Wired internet only

C. 5G and future 6G networks

Expected

D. Infrared telephony

FEEDBACK

Solution : Wireless tech like 5G and low-power communication make embedded systems more capable and connected. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=863

QUESTION 8

How have embedded systems improved user experience in recent years?

Not Attempted **Score: 0 / 1**

Options

A. They run operating systems on mainframes

B. They reduce need for user interaction

Expected

C. They enable adaptive behavior like auto power control

D. They increase device size and cost

FEEDBACK

Solution : Embedded systems improve usability by making devices smarter—e.g., adjusting power based on food in a microwave. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=1266

QUESTION 9

Which of the following best describes a general-purpose computer?

Not Attempted **Score: 0 / 1**

Options

A. It performs only one fixed task

B. It is embedded inside a larger device

Expected

C. It runs many applications with flexible input/output

D. It cannot be programmed by users

FEEDBACK

Solution: General-purpose computers like laptops can run many applications and accept a wide range of inputs. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=1305

QUESTION 10

What is a key challenge while developing embedded systems?

Not Attempted **Score: 0 / 1**

Options

A. Excess memory causing overflow

B. Too many user interfaces to manage

Expected

C. Limited processing power and memory

D. Lack of network connectivity

FEEDBACK

Solution : An embedded system is a computer system that does a particular task inside a machine or larger electrical system. Youtube link : https://youtu.be/3cqu_xnuPnw?feature=shared&t=1922